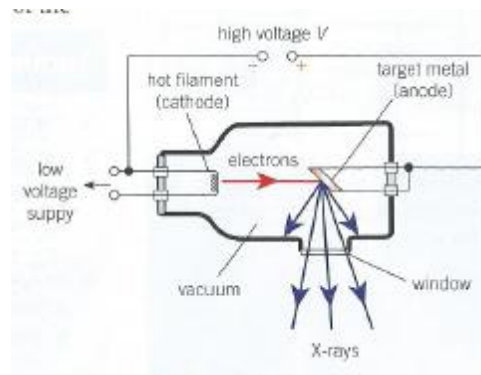


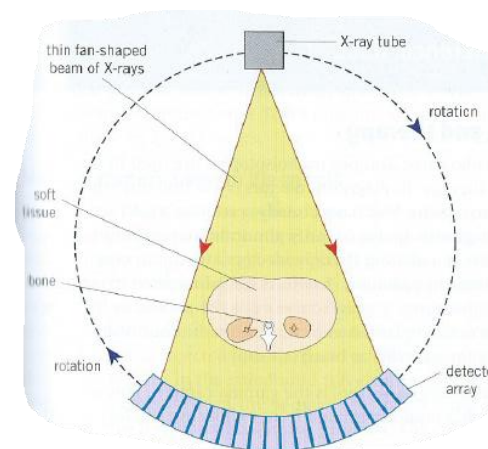
General	
Therapy	Treatment of a patient: eg radiotherapy is using radioactivity to treat a disorder such as cancer
Diagnosis	Trying to find out what is wrong: all of the imaging techniques described below are used for diagnosis.
examples	Using a gamma source used to kill specific cells – eg cancer cells – would be a therapy . Using x-rays or gamma rays to create an image is a diagnostic tool.
Attenuation	Attenuation is the reduction of a quantity as it passes through a medium (any quantity, not just x-rays). Eg light is attenuated as it passes through water – ie the deeper the water, the less light penetrates.

1 X-rays	
X-rays	High energy electromagnetic radiation, only gamma rays have higher energy and wavelength. Gamma rays are produced in the nucleus, X-rays are not
X-ray tube	Used to produce X-rays, by electrons striking a metal target (anode)
	The electrons are produced from a hot filament – this is thermionic emission , just like in an electron gun Electrons strike a rotating anode , and X-rays are produced



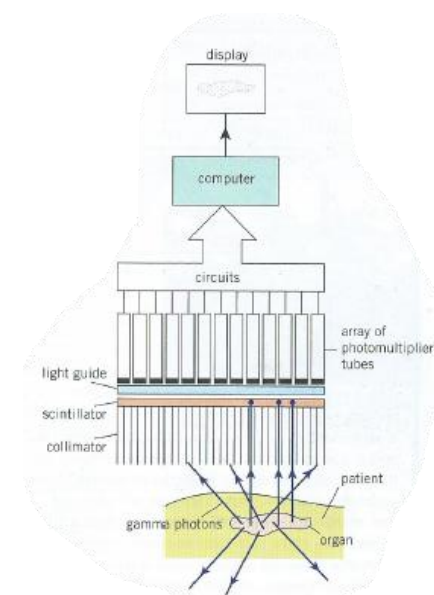
2. Interaction of X-rays with matter	
Attenuation and x-rays	Bone attenuates x-rays more than soft tissue; this fact is used to produce x-ray images
There are 4 ways in which attenuation happens	• Simple scattering • Photoelectric effect • Compton scattering • Pair production
Attenuation coefficient	The transmitted intensity is given by $I = I_0 e^{-\mu x}$ I_0 is the initial intensity μ is the attenuation coefficient (unit m^{-1}) x is the thickness
Contrast medium	A substance with a high attenuation coefficient (good at blocking x-rays) can be added to soft tissue to enable a good x-ray image. Eg Iodine injected into blood will block x-rays and blood vessels will show up in the image

3. CAT scans	
Computerised Axial Tomography	A sophisticated use of x-rays used to build up an image
Axial	x-rays are directed at a patient in a narrow 'fan', which is rotated around the patient (the 'axis'). Detectors pick up the x-rays which pass through.
Computerised	The scanning process will produce a complex set of readings. These need to be processed to form an image
Tomos	Tomos means 'slice' – the images produced are slices, or cross-sections through the patient. A series of these images is often taken, of parallel slices



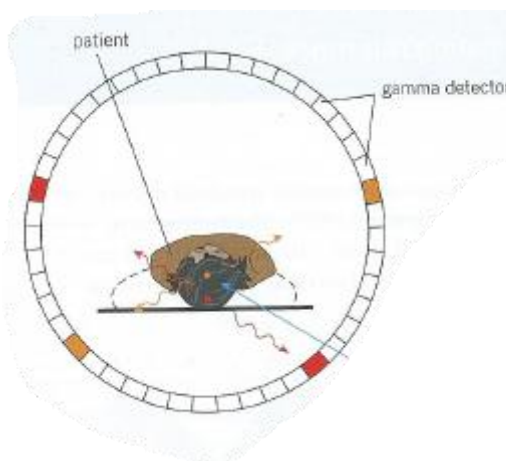
4. The gamma camera	
Method	A gamma-emitting radioisotope is used as a tracer: introduced into the body and allowed to circulate. A gamma camera used to see where it goes.

Radio-pharmaceuticals	For example a radioisotope might be used to make a molecule known to be stored in the kidneys.
Tracers	The radioisotope used must be a gamma emitter with a short half life
Collimator	Device to align a beam in a specific direction – here, lead is used to align the gamma rays perpendicular to the photomultiplier tubes
Scintillator	A material that emits light when struck by ionising radiation (gamma in this example). Also used in Rutherford's experiment, 24.1
Light guide Photomultiplier tube	The emitted photon is guided to a photomultiplier which emits an electrical pulse (a voltage pulse)



5. PET scans

Positron emission tomography	A positron emitter is used as a tracer – it is introduced into the body and allowed to circulate. The positrons annihilate with electrons immediately, producing a pair of gamma rays. The patient is surrounded by gamma detectors, which can pin-point exactly where the gamma rays came from – and therefore where the tracer element is.
Tracer	F-18 is used, half life 110 minutes
Expensive	Because the F-18 has such a short half life it must be produced on-site – with a particle accelerator



6. Ultrasound

Ultrasound	Sound waves with frequency above 20kHz
Method	Ultrasound waves are produced, transmitted into a body, where they are reflected at boundaries (eg the surface of an organ). The reflected ultrasound is analysed to create an image
Transducer	A crystal which produces ultrasound from an electrical signal, and then receives the reflected ultrasound signal and turns it back into an electrical signal for analysis. Uses the piezoelectric effect
Piezo-electric effect	Some crystals change shape when an emf is applied; if a high frequency emf is applied the ultrasound waves are produced. Works in reverse: returning ultrasound makes the crystal shape oscillate, which produced an emf
A type ultrasound scan	A one-dimensional scan, used to determine distances – very like sonar.
B type ultrasound scan	The familiar type as used for an unborn baby, produces a two-dimensional image

7. Acoustic impedance

Definition	The acoustic impedance of a material is $Z = \rho c$ where ρ is the density of the material and c is the speed of sound in that material ($\text{kgm}^{-2}\text{s}^{-1}$)
Reflection and transmission	Ultrasound is reflected from boundaries between substances – the greater the <i>difference</i> in acoustic impedance, the greater the amount reflected.

Intensity reflection coefficient (I_r/I_o)

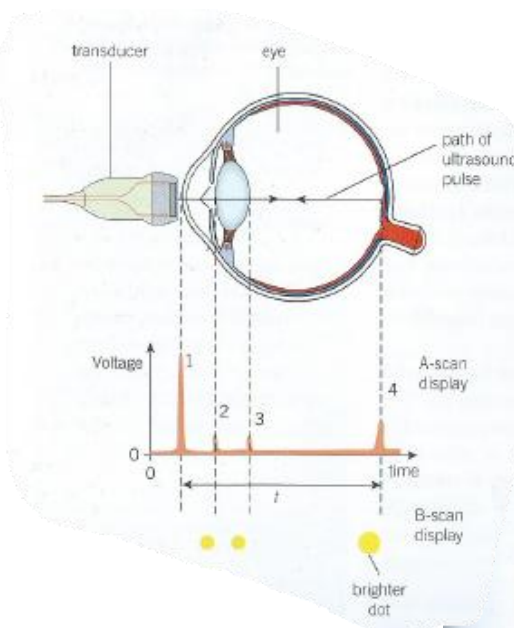
I_r/I_o is a value between 0 and 1; it is the fraction of the intensity reflected at a boundary between materials 1 and 2

Intensity reflection coefficient (I_r/I_o)

$$\frac{I_r}{I_o} = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$$

Coupling / impedance matching / acoustic matching

To maximise transmission of ultrasound we use a gel of Z value similar to the patient's skin, so that I_r/I_o is close to 0.



8. Doppler imaging

Purpose	This method allows us to detect movement of substances within the body, not just their positions.
Ultrasound and speed	The frequency of the returning ultrasound beam is altered if it is reflected off a moving object. This change in frequency is proportional to the speed of movement
Doppler effect	This change of frequency is the same as the doppler effect covered at GCSE, the only difference is that the sound is reflected off the moving object, rather than being emitted by it.

$$\Delta f = \frac{2fvcos\theta}{c}$$

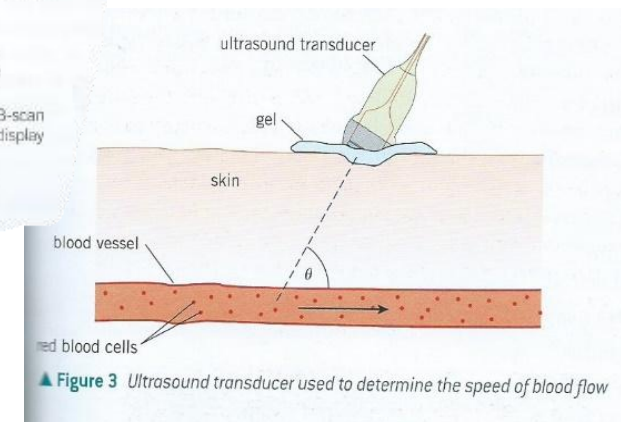
Δf is the change in frequency

f is the frequency

v is the speed of the moving object

c is the speed of the ultrasound in the material,

θ is the angle between the ultrasound beam and the direction of movement



▲ Figure 3 Ultrasound transducer used to determine the speed of blood flow